This assignment will not be graded and is only for practice.

Level 1:

1) Let $Ax = b$ be a matrix representation of a system of linear equations. Let $[R|c]$ be the reduced row echelon of the augmented matrix $[A|b]$ corresponding to the system. Choose the set of correct options. ***1 point***

[Hint: Recall reduced row echelon form.]

- [ ] If the system $Rx = c$ has infinitely many solutions, then the system $Ax = b$ has infinite solutions.
- [ ] If the system $Rx = c$ has no solutions, then the system $Ax = b$ has a unique solution.
- [ ] If the system $Rx = c$ has a unique solution, then the system $Ax = b$ has no solution.
- [ ] If the system $Rx = c$ has a unique solution, then the system $Ax = b$ has a unique solution.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If the system $Rx = c$ has infinitely many solutions, then the system $Ax = b$ has infinite solutions.
If the system $Rx = c$ has a unique solution, then the system $Ax = b$ has a unique solution.

Consider a system of linear equations
$$\begin{aligned} 2x_1 + x_2 &= 1 \\ -x_1 + x_3 + x_4 &= -1 \\ x_1 + x_2 - x_3 + x_4 &= 2 \\ -x_1 + x_3 + x_4 &= 1. \end{aligned}$$
If the following matrix represents the augmented matrix of the system, then answer the questions 2,3 and 4.

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & a_{13} & a_{14} & b_1 \\ a_{21} & a_{22} & a_{23} & a_{24} & b_2 \\ a_{31} & a_{32} & a_{33} & a_{34} & b_3 \\ a_{41} & a_{42} & a_{43} & a_{44} & b_4 \end{array}\right]$$

2) Find the value of a_{22}.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

3) Find the sum of the elements of the 4-th row of the augmented matrix.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

4) Find the sum of the elements of the 3-rd column of the augmented matrix.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

5) Let $Ax = b$ be a matrix representation of a system of linear equations and $b = 0$. Choose the set of correct options. *1 point*

[Hint: Recall the definition of the trivial solution of a system of linear equations and applicability of Gauss Elimination Method.]

- [ ] If A is an invertible matrix then the system has no solution.
- [ ] If A is an invertible matrix then the system has a unique solution.
- [ ] If A is an invertible matrix then the trivial solution is the only solution for the system.
- [ ] If $det(A) = 0$, then the system has infinitely many solutions.

No, the answer is incorrect.
Score: 0

Accepted Answers:

If A is an invertible matrix then the system has a unique solution.
If A is an invertible matrix then the trivial solution is the only solution for the system.
If $det(A) = 0$, then the system has infinitely many solutions.

6) Let $Ax = b$ be a matrix representation of a system of linear equations and $b = 0$. Choose the set of correct options. *1 point*

- [ ] If A is an invertible matrix, then the system has no solution.
- [ ] If A is an invertible matrix, then the system has a unique solution.
- [ ] If A is an invertible matrix, then the trivial solution is a solution for the system.
- [ ] If $det(A) = 0$, then either the system has no solution or the system has infinitely many solutions.

No, the answer is incorrect.
Score: 0

Accepted Answers:

If A is an invertible matrix, then the system has a unique solution.

If $det(A) = 0$, then either the system has no solution or the system has infinitely many solutions.

7) Consider the following systems of linear equations: **1 point**

$$\text{System I:} \quad \begin{aligned} x - y &= 3 \\ -y + 2z &= 1 \\ x + y + z &= 0 \end{aligned}$$

$$\text{System II:} \quad \begin{aligned} x - y &= 3 \\ 2y + z &= 1 \\ 6y + 3z &= 0 \end{aligned}$$

$$\text{System III:} \quad \begin{aligned} x - y &= 3 \\ 2y + z &= 1 \\ 6y + 3z &= 3 \end{aligned}$$

Choose the correct option.

- [ ] All the three systems have a unique solution.
- [ ] System I has a unique solution.
- [ ] System II has no solution.
- [ ] System III has infinitely many solutions.
- [ ] Both the System II and III have infinitely many solutions.
- [ ] Both the system II and III have no solution.

No, the answer is incorrect.
Score: 0

Accepted Answers:

System I has a unique solution.
System II has no solution.
System III has infinitely many solutions.

Level 2:

8) Suppose P is a 3×3 real matrix as follows: $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

The vector X_n is defined by the recurrence relation $PX_{n-1} = X_n$. If $X_2 = \begin{bmatrix} 7 \\ 8 \\ 3 \end{bmatrix}$, what is the sum of all the elements of X_0?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 6

1 point

9) Consider the system of linear equations given below: ***1 point***

$$x - y + z = 2$$

$$x + y - z = 3$$

$$-x + y + z = 4.$$

The system of linear equations has

[Hint: Use the relation between a system of linear equations and determinant of corresponding coefficient matrix.]

- [ ] no solution.
- [ ] infinitely many solutions.
- [ ] a unique solution.
- [ ] finitely many solutions.
- [ ] 3 dependent variable.
- [ ] 2 dependent and 1 independent variable.

No, the answer is incorrect.
Score: 0

Accepted Answers:

a unique solution.
3 dependent variable.

10) Consider the system of linear equations: **1 point**

$$\begin{aligned} -x_1 + x_2 + 2x_3 &= 1 \\ 2x_1 + x_2 - 2x_3 &= -1 \\ 3x_2 + cx_3 &= d \end{aligned}$$

Choose the set of correct options.

[Hint: Recall the Gauss elimination method.]

- [ ] If $c = 2$ and $d = 1$, then the system has infinitely many solutions.
- [ ] If $c = 1$ and $d = 1$, then the system has infinitely many solutions.
- [ ] If $c = 1$ and $d = 2$, then the system has no solution.
- [ ] If $c = 3$ and $d = 2$, then the system has a unique solution.

No, the answer is incorrect.
Score: 0

Accepted Answers:

If $c = 2$ and $d = 1$, then the system has infinitely many solutions.
If $c = 3$ and $d = 2$, then the system has a unique solution.

Your score is: 0/10